

Top Agentic AI Frameworks (as of 11/8/2025, by CoPilot)

Framework	Description	Best For	Notable Features
LangGraph	Built on LangChain, uses graph-based orchestration for multi-agent workflows	Complex, stateful workflows	Deterministic execution, functional API, OpenAI-compatible, low latency [research.a...ltiple.com]
CrewAI	Role-based agent collaboration framework	Team-based agent coordination	Agents with defined roles, moderate latency, human-in-the-loop support [research.a...ltiple.com]
AutoGen	Microsoft-backed multi-agent framework	Dynamic conversations and automation	Customizable agents, seamless LLM/API integration, flexible orchestration [geeksforgeeks.org]
OpenAI Swarm	Lightweight agent orchestration using Python functions	Efficient execution with minimal LLM use	Direct function calls, low token usage, fast performance [research.a...ltiple.com]
LangChain	Foundational framework for LLM apps	General-purpose agentic workflows	Modular chains, memory, tool integration, high flexibility but higher latency [research.a...ltiple.com]
DSPy	ReAct-centric agent framework	Optimizable workflows	Evaluation-driven design, fast execution, tool abstraction [langwatch.ai]
Agno	Production-grade agent memory and abstraction	Enterprise-grade deployments	Unique memory abstractions, great documentation [langwatch.ai]
SmolAgents	Lightweight, Hugging Face/open model focused	Small-model deployments	Verbose tracing, open-model compatibility [analyticsvidhya.com]
Microsoft Semantic Kernel	SDK for integrating LLMs into apps	Enterprise AI integration	Connectors, memory, planning, and orchestration tools [turing.com]
Google ADK	Ambitious agentic infrastructure	Feature-rich setups	Rails-like vision, experimental abstractions [langwatch.ai]
MetaGPT	Multi-agent collaboration for software engineering	Code generation and team simulation	Role-based agents, GitHub integration [arxiv.org]

Agentic AI Framework Comparison (as of 11/8/2025; by ChatGPT)

Framework	Description	Best For	Notable Features
OpenAI MCP (Model Context Protocol)	Protocol that standardizes how LLMs access tools, data sources, and system capabilities through client-server interfaces.	Teaching modern agent architectures; building clean, modular tool-access pipelines; interoperable systems.	Language-agnostic; lightweight; server-based tools; strong ecosystem integration (VS Code, Cursor).
LangChain	Large modular library for chaining LLM steps, tools, retrievers, and agents.	Rapid prototyping; complex pipelines with many integrations.	Massive ecosystem; Agents/Tools/Chains; vector DB integrations; experimental but flexible.
LlamaIndex	Framework centered on data indexing and retrieval, with agents layered on top.	RAG-heavy systems; structured document workflows; instructional demos.	Index graphs; query engines; agent executors; clean RAG abstractions.
AutoGen	Multi-agent conversational framework where agents collaborate via message passing.	Research; cooperative problem solving; multi-role agent systems.	Agent-to-agent messaging; tool integration; reproducible workflows; good for demos of agent coordination.
Microsoft Semantic Kernel	Plugin-oriented AI framework with memory, planning, and orchestration.	Enterprise systems; .NET/C# applications; workflow automation.	Skills (plugins); planners; connectors; cross-language support.
CrewAI	Multi-agent teamwork framework modeled around roles and tasks.	Prototyping collaborative agent teams; creative/role-based workflows.	Task delegation; role definitions; simple orchestration; human-friendly abstractions.
Haystack Agents	Search/RAG-focused framework that includes agent execution and pipelines.	Retrieval-centric applications; QA agents; Elastic-powered workflows.	Pipelines; retrievers; agent nodes; strong search integration.
LangGraph	State-machine extension to LangChain supporting cyclic workflows and persistent agent state.	Deterministic multi-step agents; planning loops; stateful execution.	DAG/state graphs; resumable checkpoints; human-in-the-loop design.
AgentFlow	Lightweight framework for building production-grade agent workflows.	Production deployment; minimal boilerplate; observability-heavy applications.	Declarative flows; tracing; stable execution; clean API.
Transformers Agents (HuggingFace)	Python-executable tool calling using HuggingFace models.	Local experimentation; teaching tool calling basics.	Python tool execution; small but educational API; no heavy orchestration.
ReAct / Toolformer / PAL (Design Patterns)	Foundational paradigms for reasoning-act loops, tool-use labeling, and program-aided reasoning.	Teaching agent reasoning; building custom or minimal agent logic.	ReAct trajectories; self-labeled tool calls; code-execution reasoning.